Fiscal Year:	FY 2019	**Task Last Updated:**	FY 05/26/2019
PI Name:	Liu, Bin Ph.D.		
Project Title:	Three-Dimensional Dusty Plasma Experiments		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Fundamental physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	bin-liu@uiowa.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	319-335-1686
Organization Name:	University of Iowa		
PI Address 1:	Department of Physics and Astronomy		
PI Address 2:	203 Van Allen Hall		
PI Web Page:			
City:	Iowa City	**State:**	IA
Zip Code:	52242-1403	**Congressional District:**	2
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2016 Physical Sciences NNH16ZTT002N: Fundamental Physics – NASA/NSF Partnership on Science of Dusty Plasmas: Utilizing the PK-4 Facility on board the ISS
Start Date:	07/25/2017	**End Date:**	04/25/2021
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JPL
Contact Monitor:	Callas, John	**Contact Phone:**	
Contact Email:	john.l.callas@jpl.nasa.gov		
Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Goree, John Ph.D. (University of Iowa)		
Grant/Contract No.:	JPL 1579454		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	A flight experiment will be performed to study phenomena resulting from shear motion in a dusty plasma. Applying an external shear, we will probe fundamental material properties selected from these topics: lattice wave excitation and coupling, viscous transport, and microscopic motion under frictional conditions. We will take a microscopic approach to uncover the nature of macroscopic properties. We will use mesoscopic particles, which serve as analogs of molecules, but allow tracking of individual particles as they interact. The experiment will use dusty plasmas, also known as "complex plasmas." These are four-component mixtures of micron-size solid particles, electrons, ions, and neutral gas. The solid particles obtain large electric charges, so that they interact by Coulomb forces with neighboring particles. The charges are so large, that these particles behave collectively as a strongly-coupled plasma. The cloud of solid particles constitutes the sample that is studied. Microgravity enables our experiment with a three-dimensional sample. Without microgravity, under 1g conditions, sedimentation causes the dust cloud to settle into a 2D layer, or at most a few horizontal layers. Our microgravity experiment will extend the understanding of particle motion that was gained in previous ground-based experiments with smaller 2D samples. Our flight experiment will be performed using PK-4, which is a facility in the Columbus module of the International Space Station (ISS). It was built by the European Space Agency. The main diagnostic is video imaging using cameras. Image data will be analyzed to generate particle position and velocity. Physical measures that can be obtained from these image data, using methods developed for ground-based experiments, include a wave’s propagation speed and dispersion relation, temperature profile, flow velocity, or shear stress. SCIENTIFIC/TECHNICAL MERITS The experiment's goals are to answer at least one of these fundamental physics questions: (1) Can we exploit the unique properties of dusty plasma to verify the theoretical prediction of localized viscous heating in a 3D liquid? (2) Can hydrodynamic shear motion couple into the generation of longitudinal waves? (3) What is the dispersion relation of lattice waves in a liquid? (4) Does a three-dimensional dusty plasma exhibit stick-slip motion during plastic deformation? The experiment will exploit manipulation devices on PK-4 including laser manipulation. Laser light imparts momentum to the solid particles, causing them to flow through the surrounding sample region. This modulation will be a pulse that is either long or short, according to the scientific goal. The feasibility of our flight experiment is based on previous ground-based experiments and our simulation. Our laboratory experiments have confirmed localized viscous heating and lattice waves. Ground-based experiments by other scientists have also confirmed the existences of plastic flow conditions and stick-slip motion. Using a simulation under PK-4 conditions, we have established the physical mechanism of wave excitation by a pulsed shear flow. In the unlikely event that this flight experiment does not yield useful results, we have a plan for a backup experiment to investigate the superheating and supercooling phase transitions, or non-Gaussian statistics. BROADER IMPACTS This research has an overlap with statistical physics, fluid physics, and soft condensed matter physics, as well as other plasma physics areas outside of dusty plasmas. Research results will be disseminated in journals, conferences, and a website. A K-12 outreach presentation will be made at an elementary school.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research has an overlap with statistical physics, fluid physics, and soft condensed matter physics, as well as other plasma physics areas outside of dusty plasmas. Research results will be disseminated in journals, conferences, and a website. A K-12 outreach presentation will be made at an elementary school.
	During this period, we carried out a fundamental physics research to prepare for the forthcoming flight experiments: three-dimensional (3D) dusty plasma experiments using the PK-4 instrument on the International Space Station (ISS). Our research activities included flight experiment planning and flight data analysis. Progress has been made in our understanding of the microscopic and macroscopic properties of 3D dusty plasmas, at the particle level. We published in a peer-reviewed journal--our second paper that uses PK-4 flight data. For our research group, our research activities have led to six peer-reviewed journal papers and eight contributed talks. This research focused on fundamental physics at the particle level, using strongly-coupled dusty plasmas. Small particles of solid matter, also called dust, are immersed in a mixture of electrons, ions, and neutral gas, making a dusty plasma. The solid particles obtain large electric charges, so that they interact by Coulomb forces with neighboring particles. The charges are so large, that these particles behave collectively as a strongly-coupled plasma. The cloud of solid particles constitutes the sample that is studied. This dust cloud can behave like solids or liquids, allowing an investigation of a fundamental difference of solids and liquids in response to shear. Direct imaging of dust motion offers a unique opportunity to uncover the nature of the shear motion at the microscopic particle level. SHEAR MOTION AT THE PARTICLE LEVEL Shear-induced motion is of fundamental importance in understanding a wide range of phenomena including structural defects and deformations in solids, seismic waves in Earth rock movements, and shear-induced cytoskeleton remodeling in biological cells. Applying an external shear, we will probe material fundamental properties selected from these topics: lattice wave excitation and coupling, viscous transport, and microscopic motion under frictional conditions. Our flight experiment's goals are to answer at least one of these fundamental physics questions: (1) Can we exploit the unique properties of dusty plasma to verify the theoretical prediction of localized viscous heating in a 3D liquid? (2) Can hydrodynamic shear motion couple into the generation of longitudinal waves? (3) What is the dispersion relation of lattice waves in a liquid? (4) Does a three-dimensional dusty plasma exhibit stick-slip motion during plastic deformation? (5) Is there a superheated state? Can a 3D dusty plasma be supercooled? (6) Does the motion of particles deviate from Gaussian statistics, with distributions exhibiting kurtosis and skew?

Task Progress:

FLIGHT EXPERIMENT PLANNING

Our planning involved a collaboration with other institutions or agencies. Significant collaborators included PK-4 Core Team members in Germany and Russia, PK-4 Facility Science Team (FST), ESA (European Space Agency) officials, CADMOS (Microgravity and Spatial Operations Development Support Center) at the CNES (France National Center for Space Studies), and the collaborators specific to Iowa projects. Both Principal Investigator (PI) Bin Liu and Co-Investigator John Goree interacted with these collaborators frequently via email and occasionally by phone, and we also met them at various meetings and conferences. One significant meeting was the recent PK-4 workshop at the DLR (German Space Agency) in February 2019. PI Bin Liu attended this meeting, and he gave a talk regarding the progress of Iowa PK-4 projects. During this workshop, Liu also discussed our plan for the next flight experiment with Core Team scientists, on the topic of the shear motion induced by a manipulation laser.

PREPARATION FOR ISS PK-4 FLIGHT EXPERIMENT

Our proposed experiment, Pulsed Shear Motion, was selected for PK-4 Science Campaign 7, which is scheduled in Summer 2019. This flight experiment will investigate the shear motion in three-dimensional dusty plasma using the PK-4 instrument onboard the ISS.

To prepare for this forthcoming flight experiment, we designed our experiment procedure, configuration, and parameters regime, yielding a flow chart. PK-4 Core Team generated an experiment script based on our flow chart. PI Bin Liu travelled to the DLR, and tested the script using the PK-4 ground model (the apparatus was the Scientific Reference Model at the DLR). The flight experiment script is now finalized and is expected to be executed in the upcoming PK-4 Science Campaign 7.

PREPARATION FOR ESA PK-4 PARABOLIC FLIGHT CAMPAIGN

As preparation for future experiments on the ISS, we will participate in the ESA PK-4 Parabolic Flight Campaign in May 2019. This campaign will use the PK-4 flight model at University of Giessen. Our German collaborators at University of Giessen will perform our parabolic flight experiment. We planned fifteen parabolas to investigate pulsed shear motion, including topics such as localized viscous heating, waves excited by pulsed shear flow, and plastic deformation and yielding.

To prepare for this parabolic flight campaign, PI Bin Liu traveled to University of Giessen and worked with scientists at Giessen, yielding flow charts and preliminary experiment scripts for Iowa experiments.

PARTICIPATION OF PK-4 SCIENCE CAMPAIGN #5

PI Bin Liu participated in PK-4 Science Campaign #5 at Toulouse France in November 2018. During his week-long stay in Toulouse, PI Bin Liu participated in the execution of Iowa's first flight experiment at the ISS, and he also observed all other PK-4 flight experiments performed in the Campaign 5.

PK-4 ISS FLIGHT DATA

Research activities on-site at Iowa also included a survey of the flight data from previous missions. During this report period, we completed an analysis of the flight data from Science Campaign #2, yielding a paper published in a peer-reviewed journal. This is our second paper that uses PK-4 flight data, for the project on three-dimensional (3D) dusty plasma experiments.

During the past year, the Iowa group received the flight data from Science Campaigns 4 and 5. We also successfully obtained the PK-4 ground data including the quick-look data for previous missions. We have so far surveyed the data for most missions. To expedite our data analysis, we also hired one undergraduate, who was funded by the National Science Foundation (NSF).

OBSERVATION OF PARTICLE DIFFUSION IN A 3-D CLUSTER IN PK-4

We found that the random motion of particles in a 3D dusty plasma under microgravity conditions can exhibit diffusive motion, with nearly Gaussian statistics. This work was a continuation of our work for the study of statistical physics and transport at the particle level, regarding the fundamental physics questions related to Goal #6 in our project proposal.

Our finding is based on an analysis of the PK-4 flight data from a previous mission of the PK-4 instrument onboard the International Space Station. For this work, we carried out a survey of PK-4 flight experiments to identify an experimental run where random motion was the dominant motion for individual dust particles. We identified three experimental runs in Science Campaign 2, from the same day, June 16, 2016, which were all performed under similar conditions and with similar results. A three-dimensional (3D) cluster of dust particles was trapped by a RF coil at 81.36 MHz, in a neon plasma at 60 Pa. The dust particles were 1.3 micron-diameter melamine formaldehyde (MF) microspheres. They formed a cloud that remained in a stable location, with a small hydrodynamic flow velocity of about 0.08 mm/s. These dust particles were tracked using video microscopy in a cross section of the 3D dust cloud. Images of particle motion were captured at a frame rate of 35 Hz. The mean-square displacement (MSD), the probability distribution function (PDF) of displacement, and the diffusion coefficient were determined.

Our finding is reported in a paper published in a peer-reviewed journal. In this paper, which is our second publication using PK-4 flight data, we reported our experimental finding of diffusive motion and nearly Gaussian distribution for the random motion of dust particles. We also determined a diffusion coefficient, which is 0.0017 mm2/s. Another measure of random motion, the probability distribution function (PDF) of particle displacement, fits a Kappa distribution with Kappa = 6.9, indicating that the random motion was nearly Gaussian. In addition to these experimental findings, we also reported a method of correcting particle mean-square displacement (MSD) data to account for a small hydrodynamic flow.

BROADER IMPACTS

Our research results have been disseminated in journals and at conferences. Our research group has published six papers in peer-reviewed journals and given seven contributed talks in various conferences.

Bibliography Type:	Description: (Last Updated: 10/05/2022)
Abstracts for Journals and Proceedings	Liu B, Goree J. "Shear deformations in dusty plasma." Presented at the 15th Dusty Plasma Workshop, Baltimore, MD, May 29-June 1, 2018. Abstracts. 15th Dusty Plasma Workshop, Baltimore, MD, May 29-June 1, 2018. , Jun-2018
Abstracts for Journals and Proceedings	Wei Z, Liu B, Goree J, Pustylnik MY, Thomas HM, Fortov VE, Lipaev AM, Usachev AD, Molotkov VI, Petrov OF, Thoma MH. "Diffusive motion in a three-dimensional cluster in PK-4." Presented at the 15th Dusty Plasma Workshop, Baltimore, MD, May 29-June 1, 2018. Abstracts. 15th Dusty Plasma Workshop, Baltimore, MD, May 29-June 1, 2018. , Jun-2018
Abstracts for Journals and Proceedings	Wong CS, Goree J, Liu B. "Fluctuation theorem confirmed in a dusty plasma." Presented at the 15th Dusty Plasma Workshop, Baltimore, MD, May 29-June 1, 2018. Abstracts. 15th Dusty Plasma Workshop, Baltimore, MD, May 29-June 1, 2018. , Jun-2018
Abstracts for Journals and Proceedings	Sen A, Ganguli G, Crabtree C, Goree J, Liu B, Tiwari S. "Experimental observation of cnoidal wave structures of dust acoustic waves." Presented at the15th Dusty Plasma Workshop, Baltimore, MD, May 29-June 1, 2018. Abstracts. 15th Dusty Plasma Workshop, Baltimore, MD, May 29-June 1, 2018. , Jun-2018
Articles in Peer-reviewed Journals	Wei Z, Liu B, Goree J, Pustylnik MY, Thomas HM, Fortov VE, Lipaev AM, Usachev AD, Molotkov VI, Petrov OF, Thoma MH. "Diffusive motion in a 3D cluster in PK-4." IEEE Transactions on Plasma Science. 2019 Jan; 7 pages. Early access January 31, 2019. https://doi.org/10.1109/TPS.2019.2893155 , Jan-2019
Articles in Peer-reviewed Journals	Liu B, Goree J, Flanagan T, Sen A, Tiwari S, Ganguli G, Crabtree C. "Experimental observation of cnoidal waveform of nonlinear dust acoustic waves." Physics of Plasmas. 2018 Nov;25(11):113701. https://doi.org/10.1063/1.5046402 , Nov-2018